Information Engineering“IE —INT 461

9 Project Management

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Outline

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4. Displaying the Work Breakdown Structure
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   - What Are the Main Types of Task Patterns?
   - STEP 3: CALCULATE THE CRITICAL PATH
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The objective of this Chapter

- Explain Project planning techniques, scheduling, monitoring,and reporting.
- Describe work breakdown structures; task patterns, and critical path analysis
- Explain techniques for estimating task completion times and costs
- Describe various scheduling tools, including Gantt charts and PERT/CPM charts
- Analyze task dependencies, duration's, start dates, and end dates
- Describe project management software and how it can assist you in project planning, estimating, scheduling, monitoring, and reporting
- Discuss the importance of project risk management
- Understand why projects sometimes fail

Mission of Project Manager and software engineer

- Project planning includes identifying all project tasks and estimating the completion time and cost of each.
- Project scheduling involves the creation of a Specific timetable, usually in the form of charts that show tasks, task dependencies, and critical tasks that might delay the project. Scheduling also involves selecting and staffing the project team and assigning specific tasks to team members. Project scheduling uses **Gantt charts and PERT/ CPM charts**, which are explained in the following sections.
- Project monitoring requires guiding, supervising, and coordinating the project team's workload. The project manager must monitor the progress, evaluate the results, and take corrective action when necessary to control the project and stay on target.
- Project reporting includes regular progress reports to management, users, and the project team itself. Effective reporting requires strong communication skills and a sense of what others want and need to know about the project.

Project Activities and Planning Steps

- Step 1: Create a Work Breakdown Structure.
  - What Is a Gantt Chart?
  - What Is a PERT/CPM Chart?
  - Which Type of Chart Is Better?
  - Identifying Tasks in a Work Breakdown Structure
- Step 2: Identify Task Patterns.
  - What Are Task Patterns?
  - How Do I Use Task Boxes to Create a Model?
  - What Are the Main Types of Task Patterns?
  - How Do I Identify Task Patterns?
  - How Do I Work With Complex Task Patterns?
- Step 3: Calculate the Critical Path.
  - What Is a Critical Path?
  - How Do I Calculate the Critical Path?

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Create a Work Breakdown Structure

- On any given day, a project manager might perform one or more of the activities listed above (MISSION). However, as Figure 1 suggests, each activity is part of a larger framework, which includes three key steps in project planning:
  - Create a work breakdown structure.
  - Identify task patterns.
  - Calculate the critical path.

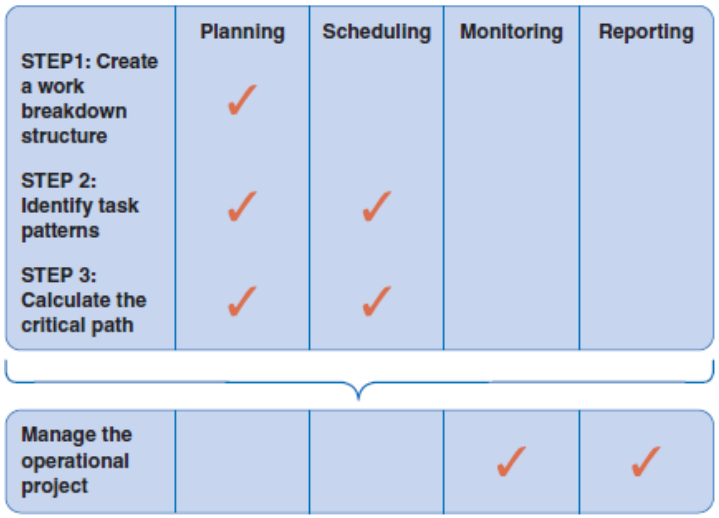


Figure: Planning Phase

Create a Work Breakdown Structure (WBS)

- A **WBS** must clearly identify each **task and include an estimated duration**.
- **A task, or activity,** is any work that has a beginning and an end and requires the use of company resources such as **people**, **time**, or **money**.
- Examples of tasks include conducting interviews, designing a report, selecting software, waiting for the delivery of equipment, or training users.
- Tasks are basic units of work that the project manager plans, schedules, and monitors — so they should be relatively small and manageable.
- In addition to tasks, every project has **events**, or **milestones**.
- An event, or milestone, is a recognizable reference point that you can use to monitor progress.

The difference between Events and milestone

For example, an event might be the start of user training, the conversion of system data, or the completion of interviews. A milestone such as “Complete 50 percent of program testing" would not be useful information unless you could determine exactly when that event will occur.

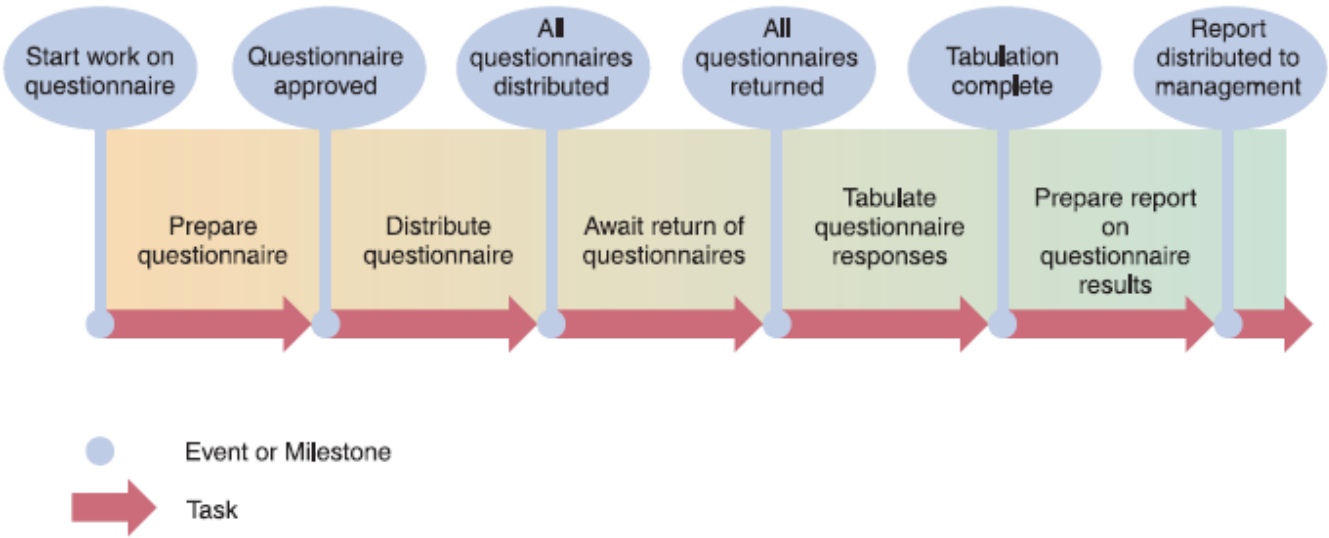


Figure: Events and milestone

Listing the Tasks

First version

First, reserve the meeting room. Then order the marketing materials and brief the managers. After the briefings, send out customer e-mails and burn sample DVDs. When the e-mails are sent and the DVDs are ready, load the new software. When the marketing materials have arrived and the software is ready, do a dress rehearsal.

Second version

First, *reserve the meeting room*. Then *order the marketing materials* and *brief the managers*. After the briefings, *send out customer e-mails* and *burn sample DVDs*. When the e-mails are sent and the DVDs are ready, *load the new software*. When the marketing materials have arrived and the software is ready, *do a dress rehearsal*.

Third version

- First, *reserve the meeting room*.
- Then *order the marketing materials* and *brief the managers*.
- After the briefings, *send out customer e-mails* and *burn sample DVDs*.
- When the e-mails are sent and the DVDs are ready, *load the new software*.
- When the marketing materials have arrived and the software is ready, *do a dress rehearsal*.

LISTING THE TASKS While this step sounds simple, it can be challenging, because the tasks might be embedded in a document, such as the one shown in the first version of Figure 3-9. One trick is to start by highlighting the individual tasks, as shown in the second version. Adding bullets makes the tasks stand out more clearly, as shown in the third version. The next step is to number the tasks and create a table, similar to the one shown in Figure 3-10, with columns for task number, description, duration, and predecessor tasks.

Task No.	Description	Duration (Days)	Predecessor Tasks
1	Reserve the meeting room		
2	Order the marketing materials		
3	Brief the managers		
4	Send out customer e-mails		
5	Burn sample DVDs		
6	Load the new software		
7	Do a dress rehearsal		

In this table, columns have been added for task number, description, duration, and predecessor tasks.

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Estimating the duration of the task “TASK DURATION ”

A person-day represents the work that one person can complete in one day. This approach, however, can present some problems. For example, if it will take one person 20 days to perform a particular task, it might not be true that two people could complete the same task in 10 days or that 10 people could perform the task in two days.

- Some tasks can be divided evenly so it is possible to use different combinations of time and people, up to a point. For instance, if it takes two persondays to install the cables for a new local area network, one person might do the task in two days, two people in one day, or four people in half a day.
- In most systems analysis tasks, however, time and people are not interchangeable. If one analyst needs two hours to interview a user, two analysts also will need two hours to do the same interview.
- Project managers often use a weighted formula for estimating the duration of each task.

Weighted formula for estimating the duration

Project managers often use a weighted formula for estimating the duration of each task. The project manager first makes three time estimates for each task: an optimistic, or **best-case estimate** (B), a **probable-case estimate** (P), and a pessimistic, or **worst-case estimate** (W). The manager then assigns a **weight**, which is an importance value, to each estimate. The weight can vary, but a common approach is to use a ratio of B = 1, P = 4, and W = 1. The expected task duration is calculated as follows:

$$\frac{(B+4P+W)}{6}$$

For example, a project manager might estimate that a file-conversion task could be completed in as few as 20 days or could take as many as 34 days, but most likely will require 24 days. Using the formula, the expected task duration is 25 days, calculated as follows:

$$\frac{(20+(4*24)+34)}{6} = 25$$

Figure: Caption

Factors Affecting Duration

When developing duration estimates, project managers consider four factors:

- Project size
- Human resources
- Experience with similar projects
- Constraints

Displaying the Work Breakdown Structure

After you enter the task durations, the work breakdown structure will look like Figure . If you are managing a complex project with many tasks, you can use task groups, just as you would in a Gantt chart, to simplify the list. If you are using Microsoft Project, the WBS might resemble :

Task No.	Description	Duration (Days)	Predecessor Tasks
1	Reserve the meeting room	1	
2	Order the marketing materials	9	
3	Brief the managers	2	
4	Send out customer e-mails	3	
5	Burn sample DVDs	3	
6	Load the new software	2	
7	Do a dress rehearsal	1	

Task durations have been added, and the WBS is complete except for predecessor task information. The predecessor tasks will determine task patterns and

Figure: WBS done

IDENTIFY TASK PATTERNS

- In any project, large or small, tasks depend on each other and must be performed in a sequence, not unlike the commands in a software program.
- Task patterns can involve dependent tasks, multiple successor tasks, and multiple predecessor tasks.
- In larger projects, these patterns can be very complex, and an analyst must study the logical flow carefully.
- Task Boxes are used for determine the task duration/

How Do I Use Task Boxes to Create a Model?

In a PERT/CPM chart, project tasks are shown as rectangular boxes, arranged in the sequence in which they must be performed. Each rectangular box, called a task box, has five sections. Each section of the task box contains important information about the task, including the Task Name, Task ID, Task Duration, Start Day/Date, and Finish Day/Date.

TASK PATTERNS

- TASK NAME —The task name should be brief and descriptive, but it does not have to be unique in the project.
- TASK ID The task ID can be a number or code that provides unique identification.
- TASK DURATION The duration is the amount of time it will take to complete a task.
- START DAY/DAT The start day/date is the time that a task is scheduled to begin. For example, suppose that a simple project has two tasks: Task 1 and Task 2. Also suppose that Task 2 cannot begin until Task 1 is finished. An analogy might be that you cannot run a program until you turn on your computer. If Task 1 begins on Day 1 and has a duration of three days, it will finish on Day 3. Because Task 2 cannot begin until Task 1 is completed, the start time for Task 2 is Day 4, which is the day after Task 1 is finished.
- FINISH DAY/DATE The finish day/date is the time that a task is scheduled to be completed. To calculate the finish day or date, you add the duration to the start day or date. When you do this, you must be very careful not to add too many days. For example, if a task starts on Day 10 and has a duration of 5 days, then the finish would be on Day 14 — not Day 15.

DEPENDENT TASKS

- DEPENDENT TASKS When tasks must be completed one after another, like the relay race shown in Figure 3-14, they are called dependent tasks, because one depends on the other. For example, Figure 3-15 shows that Task 2 depends on Task 1, because Task 2 cannot start until Task 1 is completed. In this example, the finish time of Task 1, Day 5, controls the start date of Task 2, which is Day 6.

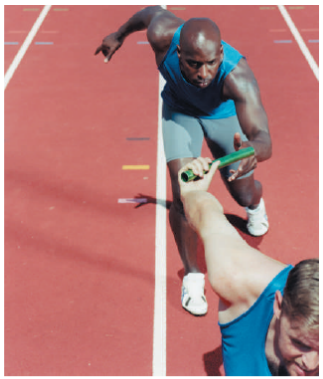


FIGURE 3-14 In a relay race, each runner is dependent on the preceding runner and cannot start until the earlier runner finishes.

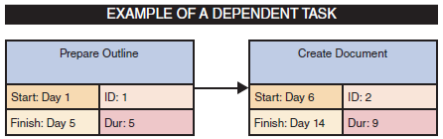


Figure: Caption

MULTIPLE SUCCESSOR TASKS

When several tasks can start at the same time, each is called a concurrent task. Often, two or more concurrent tasks depend on a single prior task, which is called a predecessor task. In this situation, each concurrent task is called a successor task. In the example shown in Figure 3-16, successor Tasks 2 and 3 both can begin as soon as Task 1 is finished. Notice that the finish time for Task 1 determines the start time for both Tasks 2 and 3. In other words, the earliest that Task 1 can finish is day 30, so day 31 is the earliest that Tasks 2 and 3 can start.

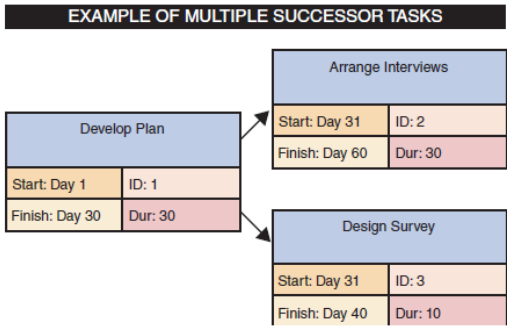


Figure: Caption

MULTIPLE PREDECESSOR TASKS

Suppose that a task requires two or more prior tasks to be completed before it can start. Figure 3-17 on the next page shows that example, because Task 3 cannot begin until Tasks 1 and 2 are both completed. Since the two tasks might not finish at the same time, the longest (latest) predecessor task becomes the controlling factor. Notice that the start for Task 3 is Day 16, not Day 6. Why is this so? Because Task 3 depends on two predecessor tasks, Tasks 1 and 2, Task 3 cannot begin until the later of those tasks is complete. Therefore, the start time for a successor task must be the latest (largest) finish time for any of its preceding tasks. In the example shown, Task 1 ends on Day 15, while Task 2 ends on Day 5, so Task 1 controls the start time for Task 3.

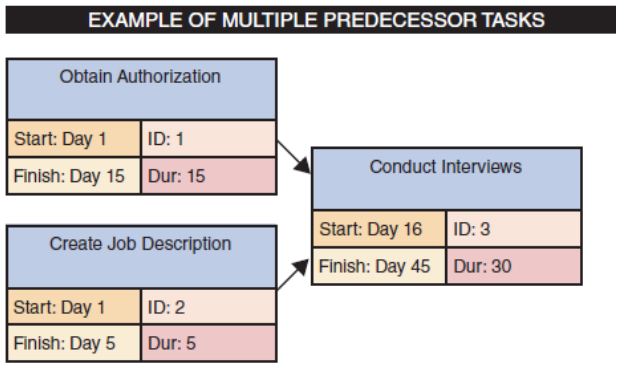


FIGURE 3-17 This example of multiple predecessor tasks shows that the start time for a successor task must be the latest (largest) finish time for any of its preceding tasks. In the example shown, Task 1 ends on Day 15, while Task 2 ends on Day 5, so Task 1 controls the start time for Task 3.

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How Do I Identify Task Patterns?

You can identify task patterns by looking carefully at the wording of the task statement. Words like **then, when, or and** are action words that signal a sequence of events. Here are three simple examples:

- Do Task 1, then do Task 2 describes dependent tasks that must be completed.
- When Task 2 is finished, start two tasks: Task 3 and Task 4 describes multiple successor tasks that can both start as soon as Task 2 is finished.
- • When Tasks 5 and 6 are done, start Task 7 indicates that Task 7 is a multiple predecessor task because it can't start until two or more previous tasks all are completed. one after the other

How Do I Work With Complex Task Patterns?

When several task patterns combine, you must study the facts very carefully to understand the logic and sequence. A project schedule will not be accurate if the underlying task pattern is incorrect. For example, consider the following three fact statements and the task patterns they represent. Examples of the task patterns are shown Figures 3-18, 3-19, and 3-20.

DEPENDENT TASKS, MULTIPLE SUCCESSOR TASKS, AND MULTIPLE PREDECESSOR TASKS Perform Task 1. When Task 1 is complete, perform Task 2. When Task 2 is finished, start two Tasks: Task 3 and Task 4. When Task 3 is complete, start two more tasks: Task 5 and Task 6. When Tasks 5 and 6 are done, start Task 7. Then, when Tasks 4 and 7 are finished, perform Task 8.

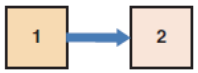


FIGURE 3-18 Dependent tasks.

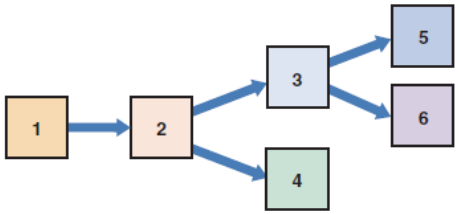


FIGURE 3-19 Dependent tasks and multiple successor tasks.

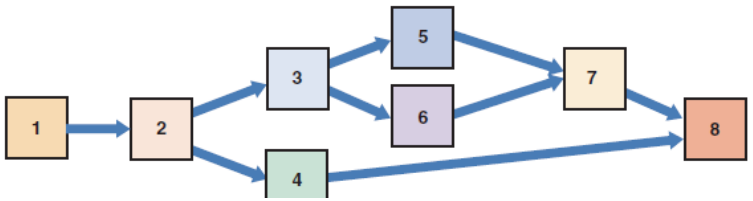


Figure Caption

a case study

CHAPTER INTRODUCTION CASE: Mountain View College Bookstore



Background: Wendy Lee, manager of college services at Mountain View College, wants a new information system that will improve efficiency and customer service at the three college bookstores.

In this part of the case, Tina Allen, systems analyst, and David Conroe, student intern, are talking about project management tools and techniques.

Participants: Tina and David
Location: Mountain View College Cafeteria, Wednesday afternoon, September 21, 2011
Discussion topics: Project planning, Gantt charts, PERT/CPM charts, Microsoft Project and Open Workbench software, project monitoring, and risk management techniques.

Tina: Hi, David. Glad I ran into you. I'd like to talk with you about project management, which we'll be using as we plan and execute the bookstore information system project.

David: *Sure. I've read a little about project management, but I don't know the specifics.*

Tina: Well, we manage business and personal projects every day, but we don't always give it much thought. To manage large-scale IT projects, you need specific tools and techniques. You also need a project manager, who is responsible for planning, leading, organizing, and controlling all the tasks.

David: *I guess that's you?*

Tina: Sure is. No matter which tools you use, the idea is to break the project down into individual tasks, determine the order in which the tasks need to be performed, and figure out how long each task will take. With this information, you can use Gantt charts or PERT/CPM charts to schedule and manage the work.

David: *I've seen Gantt charts — they're the ones that look like horizontal bar charts?*

Tina: Right. In addition to Gantt charts, we'll use PERT/CPM charts, which look like network diagrams that show all the tasks, patterns, and calculations that we'll need. We'll learn how to create PERT/CPM charts manually, and we'll also experiment with Microsoft Project and Open Workbench, which are powerful project management tools.

David: *Anything else we need to know?*

Tina: Yes. After we have a specific plan, we need to monitor it carefully, report the progress, and employ a process called risk management. If you are ready, here's a task list to get us started:

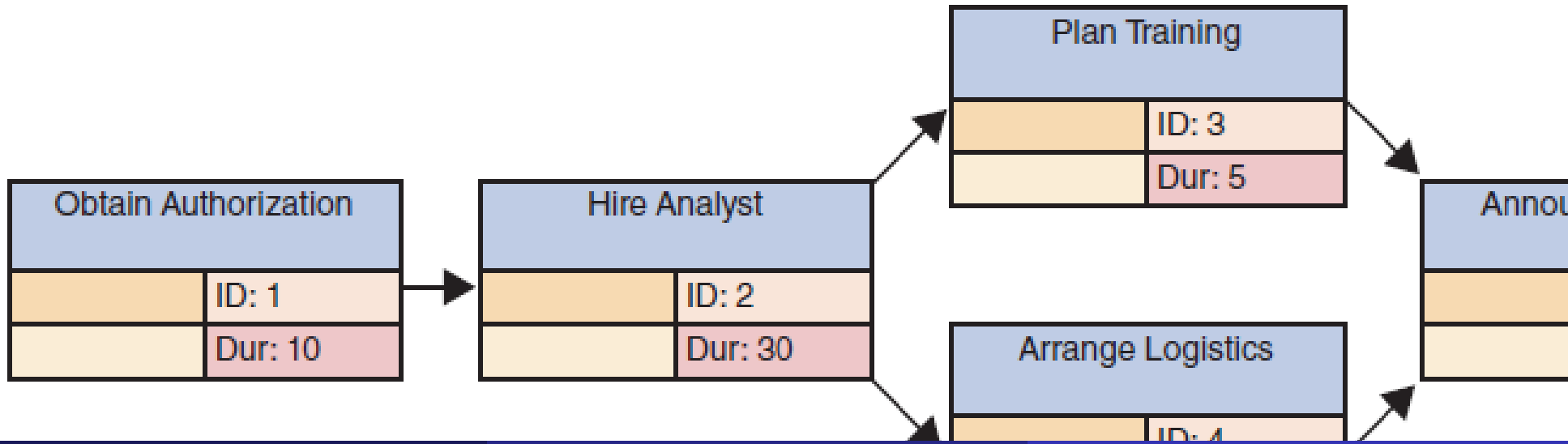
1. A critical path is a series of tasks which, if delayed, would affect the completion date of the overall project.
2. If any task on the critical path falls behind schedule, the entire project will be delayed. For example, suppose that you invite Joan and Jim to your home for dinner. Joan arrives on time, but Jim arrives 30 minutes late. Jim's arrival is part of the critical path, because you do not want to start without him, so the meal will be served 30 minutes later than originally planned. Project managers always must be aware of the critical path, so they can respond quickly to keep the project on track.
3. Microsoft Project and other project management software can highlight the series of tasks that form the critical path.

How to calculate the critical path

How Do I Calculate the Critical Path?

Figure 3-21 shows a training project with five tasks. Notice that the analyst has arranged the tasks and entered task names, IDs, and durations. First, you should review the task patterns. In this example, Task 1 is followed by Task 2, which is a dependent task. Task 2 has two successor tasks: Task 3 and Task 4. Tasks 3 and 4 are predecessor tasks for Task 5.

The next step is to determine start and finish dates, which will determine the critical path for the project. The following explanation will guide you through a step-by-step process. The result is shown in Figure 3-22 on the next page.



Identify the critical path

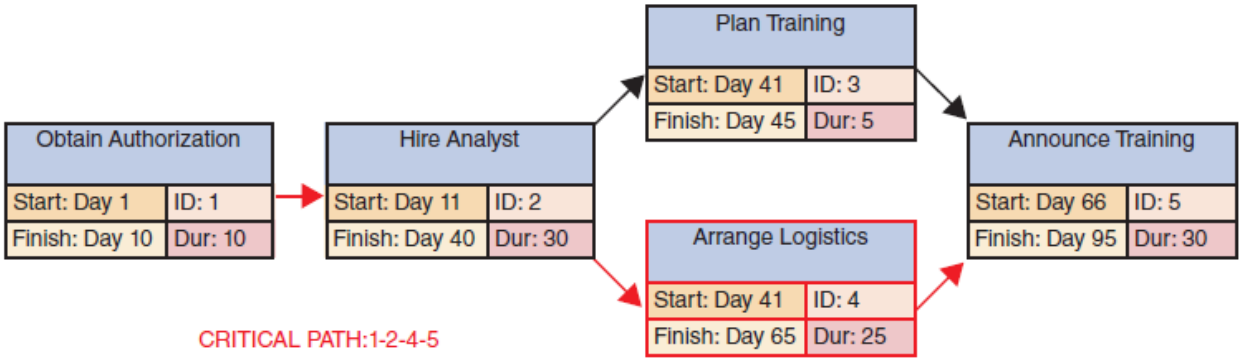


FIGURE 3-22 Now the analyst has entered the start and finish times, using the rules explained in this section. Notice that the overall project has a duration of 95 days.

- Task 1 starts on Day 1 and has a duration of 10 days, so the finish date is Day 10.
- Task 2, which is dependent on Task 1, can start on Day 11 — the day *after* Task 1 ends. With a duration of 30 days, Task 2 will end on Day 40.
- Tasks 3 and 4 are multiple successor tasks that can start after Task 2 is done. Task 2 ends on Day 40, so Tasks 3 and 4 *both* can start on Day 41. Task 3 has a duration of 5 days, and will end on Day 45. Task 4 has a duration of 25 days, and will not end until Day 65.
- Task 5 depends on Tasks 3 and 4, which are multiple predecessors. Because Task 5 depends on *both* tasks, it cannot start until the later of the two tasks is complete. In this example, Task 3 ends earlier, but Task 4 will not be completed until Day 65, so Task 5 cannot start until Day 66.

Recall that the critical path is a series of tasks which, if delayed, would affect the final completion date of the overall project. In this example, Tasks 1 and 2 are the first tasks on the critical path. Now look at Task 5, which cannot start until both Tasks 3 and 4 are done. In this case, Task 4 is the controlling factor, because Task 4 finishes on Day 65, which is 20 days later than Task 3, which is completed on Day 45. Therefore, the start date for Task 5 is determined by the finish date for Task 4. In contrast, Task 3 has slack time, and could be delayed up to 20 days without affecting Task 5. Slack time is the amount of time that the task could be late without pushing back the completion date of the entire project. Tasks 1, 2, 4, and 5 represent the critical path, which is high-

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EXAMPLE

Task No.	Description	Duration (Days)	Predecessor Tasks
1	Develop Plan	1	-
2	Assign Tasks	4	1
3	Obtain Hardware	17	1
4	Programming	70	2
5	Install Hardware	10	3
6	Program Test	30	4
7	Write User Manual	25	5
8	Convert Files	20	5
9	System Test	25	6
10	User Training	20	7, 8
11	User Test	25	9,10

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SOLUTION

STEP 1: CREATE THE WORK BREAKDOWN STRUCTURE

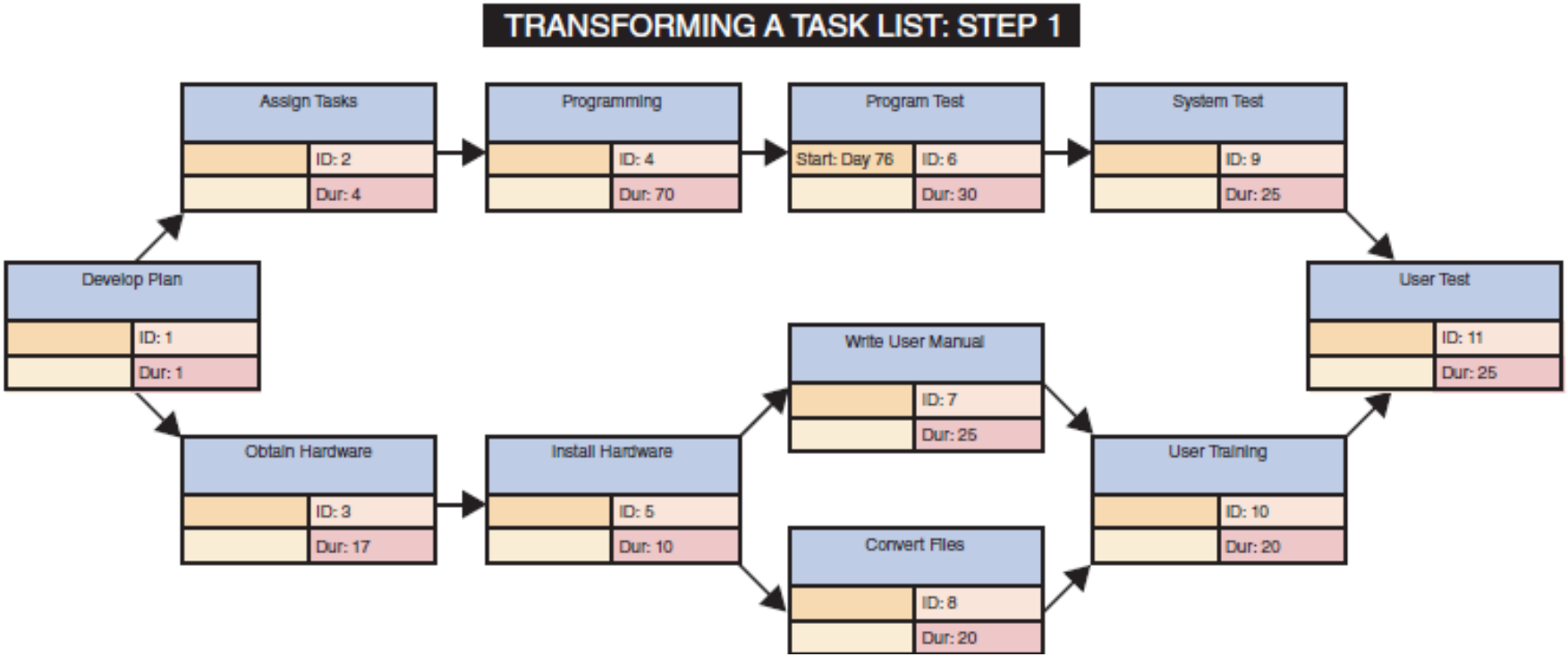


Figure: WBS

The project will be completed on Day 155. Also, you note that Tasks 1, 2, 4, 6, 9, and 11 represent the critical path shown by the red arrows.

ENTER START AND FINISH TIMES 1

there are three important rules you must keep in mind:

- Rule 1 —If a successor task has more than one predecessor task, use the latest finish time of the predecessor tasks to determine the start time for the successor task.
- Rule 2 —If a predecessor task has more than one successor task, use the predecessor task's finish time to determine the start time for all successor tasks.
- Rule 3 —Continuing from left to right, add the task duration for each task to its start time to determine and enter its finish time. Again, be very careful not to add too many days. For example, if a task starts on Day 10 and has a duration of 5 days, then the finish would be Day 14 — not Day 15.

ENTER START AND FINISH TIMES 2

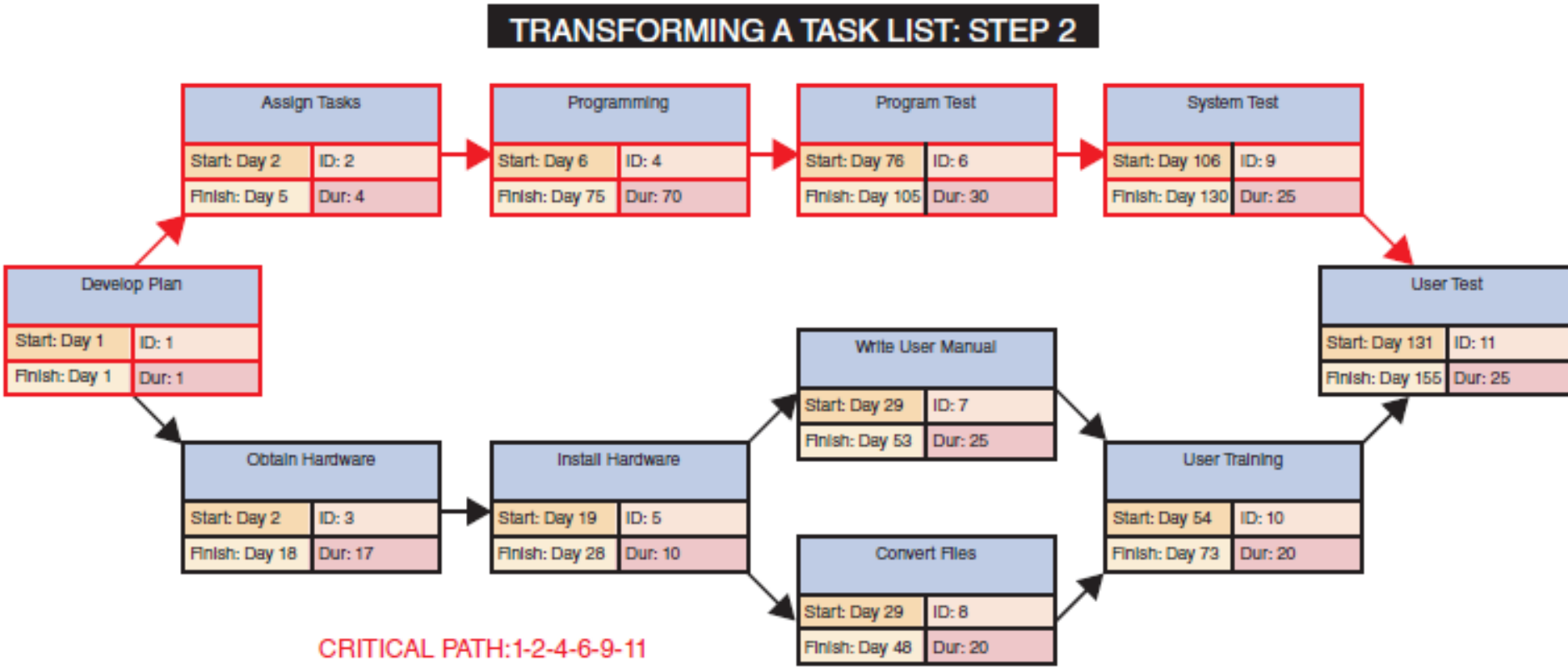


Figure: Caption

Software Driven Examples

Project managers use powerful software to help plan, schedule, monitor, and report on a project. Most project management programs offer features such as PERT/CPM, Gantt charts, resource scheduling, project calendars, and cost tracking.

- Microsoft Project management
- Open Workbench, is available as free software `https://open-workbench.en.softonic.com/`

RISK MANAGEMENT

Risk Management

Every IT project involves risks that systems analysts and project managers must address. A risk is an event that could affect the project negatively. Risk management is the process of identifying, analyzing, anticipating, and monitoring risks to minimize their impact on the project.

Steps in Risk Management

The first step in risk management is to develop a specific plan. Although project management experts differ with regard to the number of steps or phases, a basic list would include the following tasks:

- *Develop a risk management plan*. A risk management plan includes a review of the project's scope, stakeholders, budget, schedule, and any other internal or external factors that might affect the project. The plan should define project roles and responsibilities, risk management methods and procedures, categories of risks, and contingency plans.
- *Identify the risks*. Risk identification lists each risk and assesses the likelihood that it could affect the project. The details would depend on the specific project, but most lists would include a means of identification, and a brief description of the risk, what might cause it to occur, who would be responsible for responding, and the potential impact of the risk.
- *Analyze the risks*. This typically is a two-step process: Qualitative risk analysis and quantitative risk analysis. Qualitative risk analysis evaluates each risk by estimating the probability that it will occur and the degree of impact. Project managers can use a formula to weigh risk and impact values, or they can display the results in a two-axis grid. For example, a Microsoft Excel XY chart can be used to display the matrix, as shown in Figure 3-33. In the chart, notice the various combinations of risk and impact ratings for the five sample values. This tool can help a project manager focus on the most critical areas, where risk probability and potential impact are high.

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Inquiries



Assignment one

- What is project management, and what are its main objectives?
- What is the relationship between tasks and events, or milestones?
- What is a work breakdown structure? How do you create one?
- What are task patterns, and how can you recognize them?
- Compare the advantages and disadvantages of Gantt and PERT/CPM charts.
- Define the following terms: best-case estimate, probable-case estimate, and worstcase estimate, and describe how project managers use these concepts.
- How does a project manager calculate start and finish times?
- What is a critical path, and why is it important to project managers? How do you identify the critical path?
- What are some project reporting and communication techniques?
- What is risk management, and why is it important